Behavioral AI System Design – Zecpath

Objective

Design a non-invasive Behavioral AI system that analyzes observable interview behavior to assess candidate focus, engagement, and attention without relying on identity recognition or intrusive biometric processing.

System Overview

The Behavioral AI module enhances interview evaluation by monitoring behavioral signals captured through a webcam or video feed. Rather than identifying individuals, it evaluates observable interaction patterns to provide recruiters with additional context.

Core Behavioral Signals

Signal	Description
Eye Movement	Measures gaze stability and focus
Head Movement	Detects nodding, shaking, or excessive movement
Facial Engagement	Estimates attentiveness using neutral facial activity
Attention Pattern	Tracks consistency of candidate engagement

High-Level Architecture

```
Video / Webcam Feed
        │
        ▼
Face Detection Module
        │
        ▼
Eye Tracking Engine
        │
        ▼
Head Movement Analyzer
        │
        ▼
Engagement Detector
        │
        ▼
Signal Aggregator
        │
```

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        ▼
Behavior Scoring Engine
        │
        ▼
Behavior Report Generator
```

Behavioral Indicators

Focus Level

High

- Stable eye contact
- Minimal gaze shifts

Medium

- Occasional distractions

Low

- Frequent off-screen gaze

Distraction Frequency

Low

- Candidate remains focused

Medium

- Periodic distraction

High

- Frequent attention loss

Nervous Behavior

Observable signals include:

- Excessive head movement
- Repeated body gestures
- Constant posture adjustment

Engagement

High engagement indicators include:

- Active listening
- Stable facial orientation
- Consistent interaction

Non-Invasive Design Principles

- No facial recognition
- No identity tracking
- No emotion classification
- No biometric profiling
- Metadata only
- Privacy-first processing

Advantages

- Fair candidate assessment
- Privacy-preserving design
- Easy integration
- Explainable scoring

Limitations

- Camera quality affects accuracy
- Does not infer emotions
- Requires candidate consent

Future Improvements

- Gesture recognition
- Blink-rate analysis
- Adaptive attention modeling
- Real-time behavioral feedback